



Temperature-dependent charge transport and double gaussian barrier inhomogeneity in Au/PDI/n-Ge schottky diodes

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Received: 5 May 2026 / Accepted: 9 June 2026
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Abstract

An Au/PDI/n-Ge Schottky diode with a perylene diimide (PDI) organic interlayer was fabricated and its electrical behavior was investigated over the temperature range of 100–350 K. AFM analysis revealed a continuous yet non-uniform surface with nanoscale island-like features, indicating interfacial inhomogeneity. Temperature-dependent I–V characteristics exhibited strong rectification, while the barrier height increased and the ideality factor decreased with temperature, clearly deviating from ideal thermionic emission (TE). This deviation was rigorously analyzed using a double Gaussian distribution (GD) model, revealing two distinct barrier regimes associated with localized low-barrier patches embedded in a higher background barrier, consistent with inhomogeneous Schottky interfaces. The extracted mean barrier heights (~ 0.59 eV and ~ 1.004 eV) confirm spatial barrier fluctuations governing carrier transport. Modified Richardson analysis yielded values close to the theoretical constant, validating the GD-assisted TE model. Current transport analysis demonstrated a transition from ohmic conduction to trap-controlled and trap-free space-charge-limited current (SCLC), while reverse bias behavior was dominated by Schottky emission. Interface state density (N_{ss}) was found to decrease significantly with increasing temperature, indicating thermally activated passivation of interface traps. These results provide compelling evidence that charge transport in the Au/PDI/n-Ge diode is governed by Gaussian-distributed barrier inhomogeneity and interface states, highlighting the critical role of the PDI interlayer in tuning interfacial energetics and enhancing device performance.

Keywords Perylene diimide · N-Ge · Schottky barrier diodes · Barrier height inhomogeneity · Gaussian distribution

1 Introduction

Perylene diimide (PDI) and its derivatives first used as the n-type organic semiconductor material in photovoltaic devices [15] have broad prospects of application in organic electronics [73] due to its suitable optical and electronic properties, high quantum yield, high electron mobility, high molar extinction coefficient, excellent transport property, and excellent photochemical stability with strong electron affinity [26, 66, 70]. Perylene diimides (PDIs), which has a unique building block with many advantageous properties, are promising small molecules [38, 67] and have the

ability to form well ordered π - π stacking structures, which makes contribution to the electronic properties of the material [41]. All these important properties make them still attractive in many areas such as chemical/biological sensors, organic solid state lasers, organic field effect transistors (OFETs), organic solar cells, organic light emitting diodes (OLEDs), liquid-crystal displays color filters, photoconducting materials, logic gates, molecular probes, and Schottky diodes [9, 14, 29, 51, 56]. In these technological applications, especially in the electronics and optoelectronic fields, it is highly important to control the electrical properties of materials. This control mechanism can be performed by different ways and the conventional metal–semiconductor (MS) contacts for this purpose are the most commonly used method [23, 24, 37, 49, 68]. During recent years, the organic materials have been widely used between semiconductor and metal to modify the electrical properties of materials. Therefore, it is quite important to know the influence of metal/organic interfaces on the device performance. It is well known that the organic interfaces have

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a powerful effect on the device performance (Rhoderick EH. and Williams R.H. 1988, [58], and on this effect many studies have been reported [7, 8, 31, 35, 43, 47, 71]. Germanium has attracted renewed interest for next-generation electronic and optoelectronic applications owing to its high carrier mobility, narrow bandgap, and compatibility with existing semiconductor technologies [Mallikarjuna et. al 2025, Guruswathi et. al. 2026]. However, practical implementation of Ge-based devices remains challenging because of strong Fermi-level pinning at metal/Ge interfaces, which limits effective barrier height engineering and contact optimization [Kishore et. al. 2012]. Consequently, the introduction of suitable interfacial layers has emerged as a promising strategy for modifying the interfacial energetics and partially alleviating Fermi-level pinning effects [Kim et. al. 2017]. In this context, investigating PDI-based interlayers on n-Ge substrates is scientifically relevant for understanding how organic semiconductors influence barrier formation and charge transport at Ge-based contacts. Due to its high thermal stability and electron affinity, organic perylene-based materials are preferred as interfacial layers to modify the electronic properties of Schottky structures. The charge transport mechanism across such organic–inorganic heterojunctions is governed by various physical processes, where thermal activation plays a crucial role. The temperature has a significant effect on the basic characteristic properties of the diode. Particularly, the ideality factor and barrier height values exhibit a considerable change with temperature. As the temperature is increased, the ideality factor values decrease while the barrier height values increase [18, 59]. Pérez et al. [50] have investigated the change of diode parameters with temperature in a wide temperature range of 180–300 K. The ideality factor of diode fabricated from an electrospin doped polyaniline nanofiber on an n-Si substrate has shown a decrease towards high temperatures. Contrary to this decrease, the barrier height has displayed an increase. Gencer Imer and Ocak [25] have analyzed the electrical properties of a metal–semiconductor contact with organic interlayer via the I–V measurements taken in a wide temperature range. The device prepared by forming the thin film of sunset yellow dye on a p-type Si wafer has showed a good rectification property and the barrier height of device has increase together with the increasing temperature. The temperature-dependent behavior of the device barrier height has been attributed to barrier height inhomogeneity. Similar observations have been also reported by different authors in the literature.

Recently, considerable attention has been devoted to the development of stable and cost-effective organic interlayers for improving the performance of Schottky contacts. Organic semiconductors and thin interfacial layers have

been shown to effectively tune barrier height, suppress interface defects, reduce leakage currents, and improve rectification behavior while maintaining low fabrication costs. In particular, recent studies have demonstrated that properly designed organic interlayers can provide enhanced environmental stability together with improved electrical performance in metal/semiconductor structures (Huang et. al. 2019; Lin et. al. 2021). These findings further motivate the investigation of PDI-based interlayers owing to their excellent thermal, chemical, and photochemical stability as well as their favorable electron-transport characteristics.

The goal of the current work is not to assess a particular device application, even though PDI-based materials have been used in a wide range of electrical and optoelectronic devices. Rather, the goal of this work is to comprehend how a PDI interlayer alters the carrier transport mechanisms, interface states, barrier creation, and interfacial energetics in a Ge-based Schottky contact. The performance of many organic and hybrid electronic systems is largely dependent on interface-controlled charge transport, which makes this information fundamentally significant.

In this research, we prepared a Schottky diode which the perylene-diimide was used as interface on an n-type Ge wafer and later we investigated the effect of both the organic interface and the wafer on the modification of electrical properties of the prepared Schottky diode over a broad temperature range, ranging from 100 to 350 K. For this purpose, we benefited from the current–voltage measurements of the diode because the results obtained from these measurements give exhaustive information about the charge transport process and the nature of barrier occurred in the interface. The temperature dependence of the barrier height and ideality factor was analyzed using the Schottky barrier inhomogeneity model, and the presence of more than one barrier distribution in the diode was evaluated. The real barrier parameters of the diode were obtained through a modified Richardson method. We hope that this research will make an important contribution to organic electronics applications. Furthermore, rather than targeting an ideal Schottky contact, the present study aims to elucidate how a PDI interlayer influences barrier formation, interface inhomogeneity, and temperature-dependent charge transport processes at the Au/n-Ge interface. Understanding these non-ideal transport phenomena is essential for the rational design of future organic/inorganic hybrid electronic structures.

While thermionic emission models and barrier inhomogeneity have been thoroughly investigated in traditional Schottky contacts, their relevance to PDI-modified Ge interfaces is still mostly unknown. Specifically, it is still unclear how temperature-dependent transport processes, barrier height distribution, and interfacial morphology relate to each other in Au/PDI/n-Ge complexes. Therefore, by combining

temperature-dependent electrical characterisation with barrier inhomogeneity analysis over a broad temperature range, the current work seeks to provide a thorough knowledge of charge transport in this system.

2 Experimental

2.1 Device fabrication

In the study, a 500 μm thick n-type Ge (100) wafer with a resistivity of 1–10 $\Omega\text{-cm}$ was used as a substrate to prepare a perylene-diimide interfaced Schottky diode. The Ge wafer purchased from University Wafer Inc. was first chemically cleaned using the Radio Corporation of America (RCA) cleaning procedure to remove impurities from the surface [72]. Immediately after surface cleaning, high purity (99.99%) indium (In) metal was coated on the rear surface of the wafer at the 150 nm thickness in a vacuum pressure of $\sim 10^{-6}$ Torr by means of the thermal evaporation method. Then, a heat treatment on the coated wafer was carried out for 3 min at a temperature of 350 $^{\circ}\text{C}$ in vacuum to achieve a low resistance ohmic contact. Thereafter, a perylene-diimide (PDI) organic semiconductor material dissolved in toluene (at a concentration of 10 mg/mL) was coated onto the Ge wafer using Laurell Spin Coater first at a spinning rate of 500 rpm for 30 s and then at 1200 rpm for 60 s to form a PDI film. The perylene-diimide organic material used in study was synthesized following previously reported methods [36] and its chemical structure is shown in the Fig. 1. Following this, high-purity (99.999%) gold metal was evaporated onto the prepared organic film in a thermal evaporation system under a pressure of $\sim 10^{-6}$ Torr to create Schottky contacts with a diameter of 2 mm and a thickness of 150 nm (diode area (A) = 3.14 mm^2). The fabrication process and electrical measurement setup and the energy band diagram of the Au/PDI/n-Ge Schottky diode are schematically illustrated in Fig. 1a and b, respectively.

2.2 Characterization

Surface images were acquired using a Zeiss EVO scanning electron microscope. In addition, two- and three-dimensional AFM images were obtained using an Ntegra-Solaris (NT-MDT) atomic force microscope. The current–voltage (I–V) measurements of the diode were performed by a Keithley 2410 SourceMeter in the ARS Closed-Loop Cryostat (Model DE202 AI) system with the temperature controller (Lake Shore, model 331) at temperature range from 100 to 350 K.

3 Results and discussion

3.1 Surface morphology analysis by atomic force microscopy (AFM)

The surface morphology of the spin-coated PDI thin film deposited on the n-Ge substrate was investigated by AFM over a scan area of 2 $\mu\text{m} \times 2 \mu\text{m}$ using both two-dimensional (2D) and three-dimensional (3D) imaging modes (Fig. 1). The 2D AFM image reveals a continuous but non-uniform surface, where localized bright regions indicate the presence of molecular aggregation. This behavior is attributed to π – π stacking interactions inherent to perylene-based organic semiconductors.

The 3D AFM image provides a clearer visualization of the surface topography, showing randomly distributed peak–valley structures and nanoscale protrusions. These features suggest that the film does not grow layer-by-layer but rather follows an island-type growth behavior. The observed morphology is consistent with the Volmer–Weber growth mechanism, where initial nucleation is followed by the formation of isolated islands and their partial coalescence. The characteristic lateral dimensions of these surface features are estimated to be in the range of approximately 70–150 nm, while larger agglomerated structures can reach up to ~ 200 –250 nm. It should be noted that these values correspond to topographical feature sizes rather than true crystallographic grain sizes, as AFM does not directly resolve grain boundaries.

In conclusion, the AFM results indicate that the PDI film forms a continuous structure with moderate surface roughness and island-like morphology, which is expected to influence interfacial properties and charge transport behavior in the device.

The AFM observations provide additional support for the barrier inhomogeneity model proposed in this work. The island-like morphology and localized molecular aggregations observed on the PDI surface may lead to spatial variations in interfacial thickness, molecular ordering, and local electric dipole distribution. Although AFM measurements cannot directly determine local Schottky barrier heights, such nanoscale morphological fluctuations are expected to generate regions with different effective barrier heights across the Au/PDI/n-Ge interface. These regions can act as localized low-barrier and high-barrier patches, giving rise to the barrier height fluctuations described by the double Gaussian distribution model. Therefore, the morphological non-uniformity observed by AFM is consistent with the electrical evidence of barrier inhomogeneity obtained from the temperature-dependent I–V analysis.

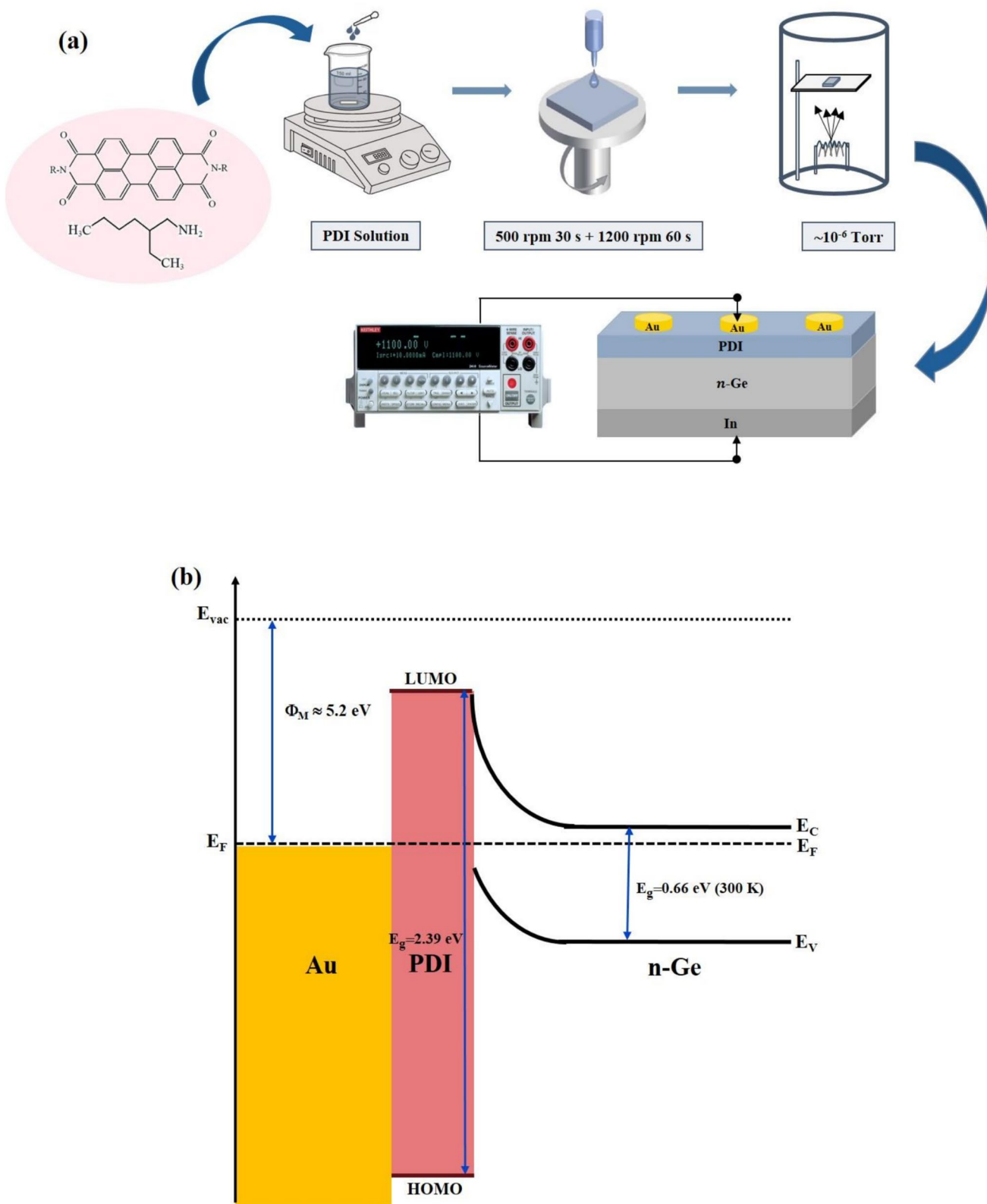


Fig. 1 **a)** Illustration of Au/PDI/n-Ge Schottky diode's production process and electrical measurement setup **b)** Energy band diagram of the Au/PDI/n-Ge Schottky diode (Ref. [34] for band gap of PDI).

3.2 Current–Voltage (I–V) characteristics

The current–voltage (I–V) measurements are the most commonly used method for determining the electrical properties of metal–semiconductor (MS) contacts and it can be easily analyzed with the help of thermionic emission (TE) theory. According to thermionic emission theory, the diode current is given as [6], [58]

$$I = I_0 \left[e^{\left(\frac{qV}{nkT} \right)} - 1 \right] \quad (1)$$

where I_0 is the saturation current given by the equation

$$I_0 = AA^*T^2 e^{\left(-\frac{q\phi_{b0}}{kT} \right)} \quad (2)$$

Also, q is the electron charge, n is the ideality factor, ϕ_{b0} is the Schottky effective barrier height at zero bias, T is absolute temperature, V is the applied bias, A is the diode area, A^* is the Richardson constant and k is the Boltzmann constant. Based on Eq. (2), the effective Schottky barrier height was calculated via Eq. (3) by utilizing the intercepts of the forward bias $\ln(I)$ - V plots at various temperatures:

$$\phi_{b0} = \frac{kT}{q} \ln \left(\frac{AA^*T^2}{I_0} \right) \quad (3)$$

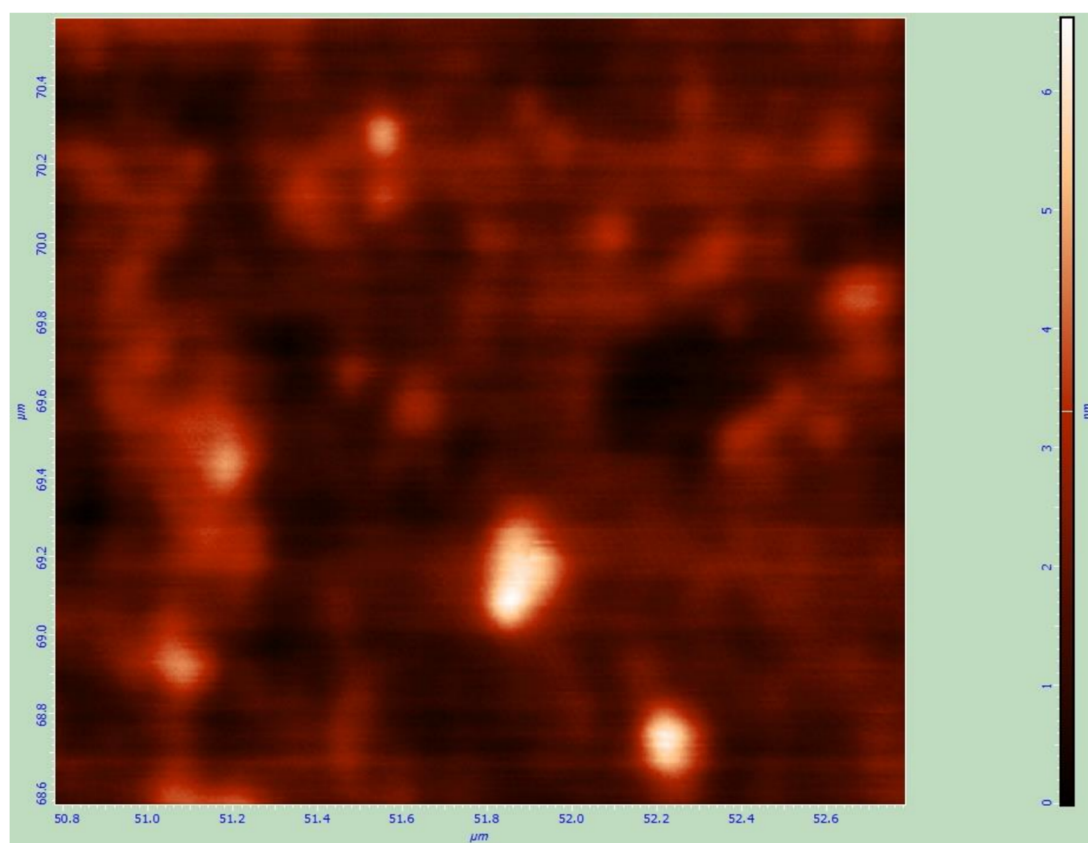
However, n can be calculated from the slope of $\ln(I)$ vs. V curve as

$$n = \frac{q}{kT} \frac{dV}{d(\ln(I))} \quad (4)$$

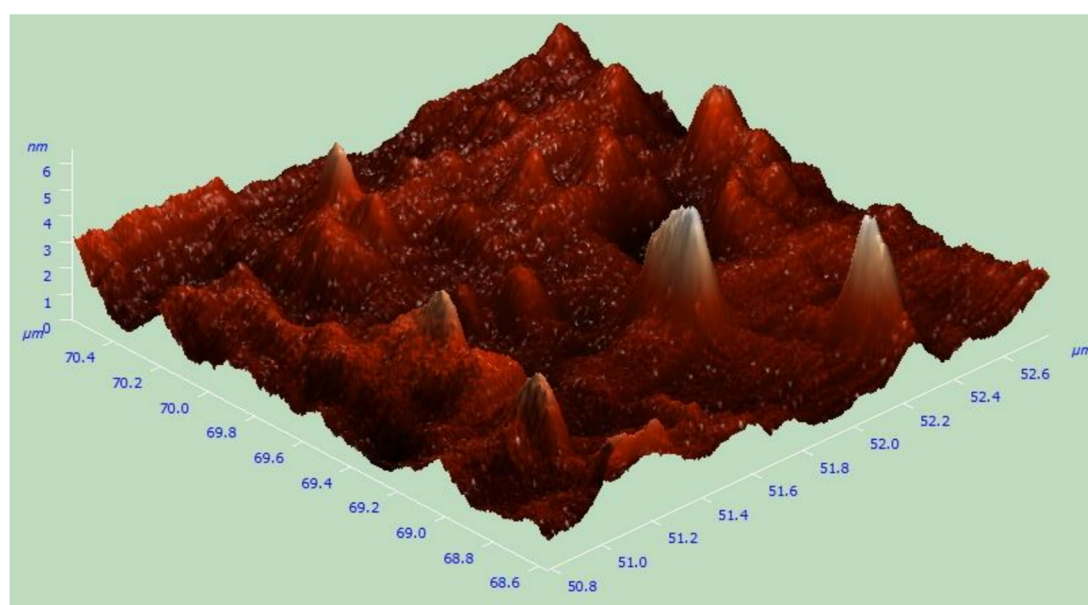
In Fig. 3, I–V and $\ln I$ - V characteristics of Au/PDI/n-Ge Schottky barrier diode was plotted for 100 K and 350 K temperature range. As illustrated in Fig. 2, the prepared Schottky contacts has been exhibited very good diode behavior at each temperature.

For each temperature value, the barrier height ϕ_{b0} and the ideality factor n of prepared diode have been determined from the forward bias current–voltage characteristics by using Eqs. (3) and (4) and the temperature dependence of barrier height and ideality factor has been shown in Fig. 4 and listed in Table 1. The diode area (A) and the effective Richardson constant (A^*) used in calculations are taken as 3.14 mm^2 and $140 \text{ A cm}^{-2} \text{ K}^{-2}$, respectively. According to the calculations, as shown in Fig. 4, the ϕ_{b0} experimental values are increased while the n values are decreased with an increase in temperature. It should be noted that the barrier heights extracted from the temperature-dependent I–V characteristics represent effective barrier heights rather than direct measurements of the semiconductor bandgap.

Therefore, a slight difference between the highest extracted barrier height and the reported bandgap energy of Ge may arise from interface-related effects such as barrier inhomogeneity, interface states, interfacial dipoles introduced by the PDI layer, and the non-ideal nature of carrier transport across the junction. This change is liner for the barrier height and it is almost exponential for the ideality factor. The linear increase in the barrier height is attributed to the increased number of electrons that surmounted the high barrier. Since charge transport across a metal–semiconductor or conducting polymer–semiconductor interface is a thermally activated process, electrons at low temperatures can overcome the lower barriers. Consequently, the current transport is primarily governed by current flowing through regions with reduced barrier height [5, 13, 58]. Briefly, since the temperature rises, the more electrons will have enough energy to exceed the high barrier, and so the dominating barrier height will increase with both temperature and bias voltage [71]. 0.611 eV barrier height value determined for the Au/PDI/n-Ge diode prepared using perylene-diimide organic interface layer is relatively higher than that achieved with traditional Au/n-Ge diode whose barrier height is 0.428 eV [10]. This organic interface layer acts as a physical spacer between the Au and n-Ge wafer, which in turn leads to a notable shift in the work function of the metal and the electron affinity of the semiconductor. In this way, the organic layer makes a difference of 0.19 eV. Although a direct Au/n-Ge reference device was not included in the present study, previously reported Au/n-Ge Schottky contacts typically exhibit barrier heights in the range of approximately 0.42–0.45 eV [Chawanda et. al. 2012, Suzuki et. al. 2014]. In comparison, the Au/PDI/n-Ge structure investigated here exhibits a noticeably higher barrier height, suggesting that the PDI interlayer modifies the interfacial energetics and barrier formation at the metal/semiconductor interface. A systematic experimental comparison between Au/n-Ge and Au/PDI/n-Ge structures would be valuable and may be considered in future studies. Besides acting as a physical spacer, the PDI interlayer may also modify the interfacial energetics through dipole formation. The insertion of the organic layer can reduce the influence of metal-induced gap states (MIGS) by partially decoupling the Au contact from the Ge surface, thereby weakening Fermi-level pinning. Furthermore, the electron-accepting nature of PDI and the charge redistribution occurring at the Au/PDI and PDI/Ge interfaces may create an interfacial dipole, leading to a shift in the local vacuum level and an increase in the effective Schottky barrier height. Therefore, the observed barrier height enhancement is likely the result of the combined action of MIGS suppression and interfacial dipole formation. Many such attempts to change the barrier height of Schottky diodes were made by forming an interfacial layer between the metal and semiconductor.



(a)



(b)

Fig. 2 Typical (a) 2D and (b) 3D AFM images of PDI thin film deposited on n-Ge

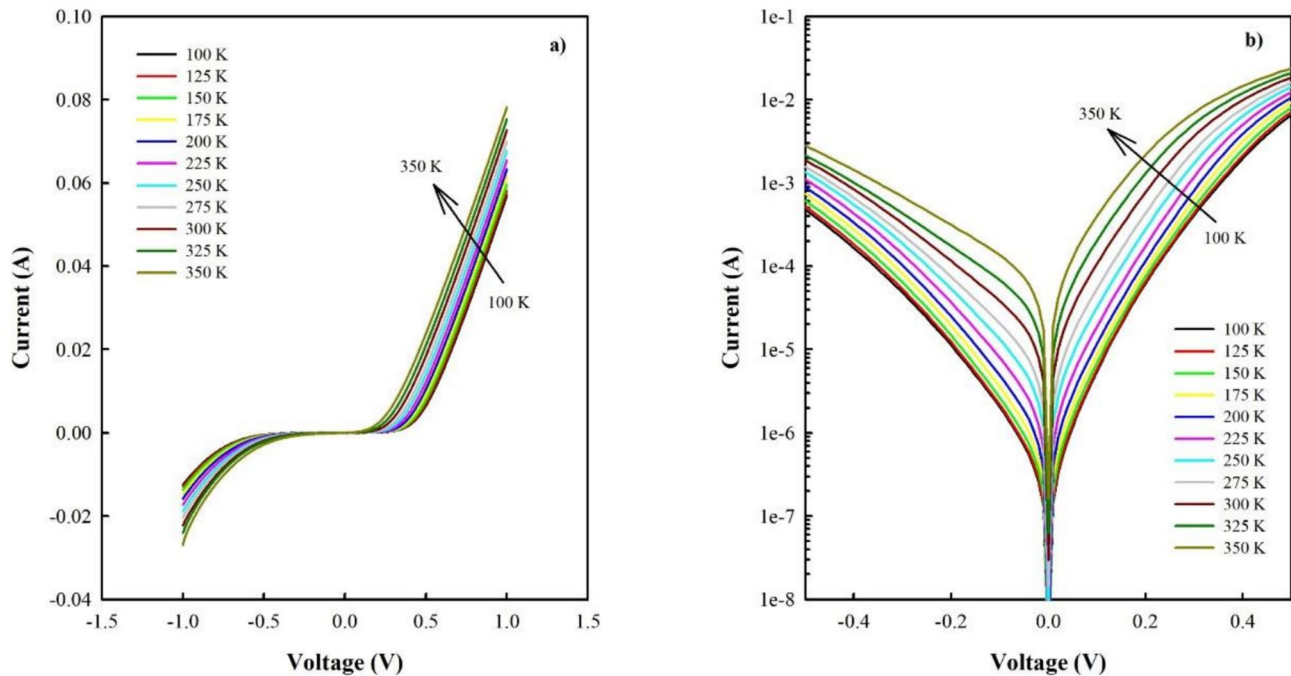


Fig. 3 (a) I-V characteristics and (b) lnI-V plots of the Au/PDI/n-Ge Schottky diode

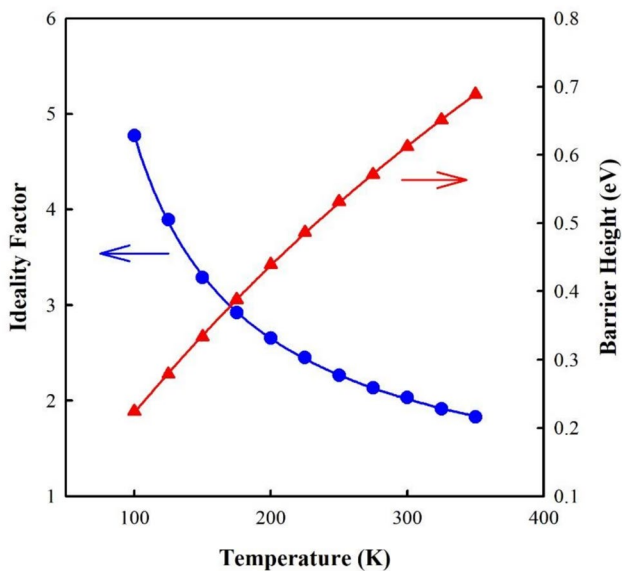


Fig. 4 The variation of n and ϕ_b values with temperature for Au/PDI/n-Ge Schottky diode

Oruç and Altındal [44] reported a barrier height of 0.52 eV for the Au/FePc/p-Si structure at room temperature. In another study, Khurelbaatar et al. [30] reported a barrier height value of 0.63 eV for the Au/n-Ge structure prepared using monolayer graphene interlayer. Despite all the studies available in the literature, to the best of our knowledge, we believe that this study will be valuable since there is no different metal contact PDI/n-Ge study in the literature.

Another important parameter in a good diode behavior is the ideality factor (n). Ideality factor is defined as a parameter that shows how much an ideal diode deviates from the ideal and the deviation in the ideality factor can be resulted from the thin interfacial layer, the interface state influences, series resistance, generation-recombination processes and image-force lowering [22]. The ideal factor value for an ideal diode is one, but this value is mostly greater than one for fabricated diode structures. In our present study, the n values for prepared Au/PDI/n-Ge Schottky diode measured at 100–350 K temperature range have been ranged from 4.77 to 1.82, respectively (Fig. 4). The ideality factor value found have decreased nearly exponentially with increasing temperature and it has been found to be greater than one for prepared Schottky diode structure. As mentioned above, obtaining high values are commonly explained by many effects such as organic layer effect, interface states, nonuniformity distribution of the interfacial charges, series resistance, generation-recombination processes, inhomogeneous film thickness, and in literature, many researcher has investigated the ideality factor values of different structures [3, 17, 21, 53]. Especially in recent years, the low ideality factors have been obtained thanks to newly developed organic semiconductors. For example, at Au/FC/n-Si structure using 4,4'-(9,9-dioctyl-9H-fluorene-2,7-diyl)bis(N,N-diphenylaniline) (FC) organic material, it has been reported by Yıldırım et al. [69] that ideality factor values vary from 1.24 at 220 K to 1.03 at 380 K, respectively. In a different

Table 1 Experimental diode parameters determined by different methods for the Au/PDI/n-Ge heterojunction across various temperatures

T	n	n Cheung	ϕ_b	ϕ_b Cheung	ϕ_b Norde	R_s	R_s Cheung (H(I)-I)	R_s Norde
(K)	$(I - V)$		$(I - V)$	(eV)	(eV)	Cheung (dV/d(lnI)-I)	(Ω)	(Ω)
100	4.77	4.72	0.224	0.225	0.216	8.315	8.315	88.119
125	3.89	3.88	0.278	0.275	0.274	8.258	8.255	81.176
150	3.28	3.26	0.333	0.342	0.331	8.237	8.229	68.543
175	2.92	2.90	0.387	0.388	0.386	8.110	8.111	47.838
200	2.65	2.64	0.439	0.443	0.439	8.024	8.027	16.441
225	2.45	2.45	0.486	0.488	0.488	7.966	7.963	11.030
250	2.26	2.26	0.531	0.533	0.534	7.910	7.911	8.563
275	2.13	2.13	0.571	0.577	0.578	7.804	7.806	7.250
300	2.03	2.02	0.611	0.612	0.616	7.739	7.741	5.758
325	1.91	1.90	0.651	0.649	0.653	7.426	7.450	6.668
350	1.82	1.82	0.688	0.681	0.687	7.308	7.309	6.529

study investigating the temperature-dependent current–voltage characteristics for Au/anthracene/n-Si diode at the 140–300 K temperature range by Kaçus et al. [27], the low n values changing from 3.00 at 140 K to 1.88 at 300 K have been observed. However, a significant influence on the electrical properties of device as well as the ideality factor and the barrier height effects is the series resistance (R_s) effect. The series resistance value can be determined analyzing the I-V characteristics of diode via Cheung and Cheung model [11]. According to Cheung and Cheung model, the forward bias current–voltage characteristics by benefiting from the thermionic emission theory is expressed as

$$I = I_0 \left[e^{\left(\frac{q(V - IR_s)}{nkT} \right)} \right] \quad (5)$$

where IR_s is the voltage drop across the series resistance of diode. When I_0 parameter is written instead of equation and V parameter is extracted from the equation, the following expression is obtained:

$$V = IR_s + n\phi_b + \frac{nkT}{q} \ln \left(\frac{I}{AA^*T^2} \right) \quad (6)$$

If this expression is derived according to I and rearranged, the Eq. (7) is found:

$$\frac{dV}{d \ln(I)} = \frac{nkT}{q} + IR_s \quad (7)$$

According to the Eq. (7) stated above, if the graph of $dV/d \ln(I)$ versus I is drawn, the resulting graph will be linear and slope of graph will give the R_s series resistance and its intercept on the current axis will give $\frac{nkT}{q}$. However, to found the ϕ_b barrier height of prepared diode, it can also be utilized Eq. (8) defined as Cheung and Cheung formula

$$H(I) = V - \left(\frac{nkT}{q} \right) \ln \left(\frac{I}{AA^*T^2} \right) = n\phi_b + IR_s \quad (8)$$

Similarly to the graph of $dV/d \ln(I)$, the plot of $H(I)$ versus I will give the ϕ_b and R_s . Figure 5 show the plots of $dV/d \ln(I)$ versus I for a wide temperature range from 100 to 350 K. As depicted in Fig. 5, the plotted curves exhibit good linearity for all temperature values. Moreover, when the temperature increases, the corresponding curves shift upward, consistent with Eq. (7). Additionally, in Fig. 6, the plot of $H(I)$ versus I has been shown for the same temperature values, and a similar behavior with increasing temperature has been seen for these drawn curves, too.

The electrical parameters obtained from Figs. 5 and 6 are given in Table 1 together with the n and ϕ_{b0} values

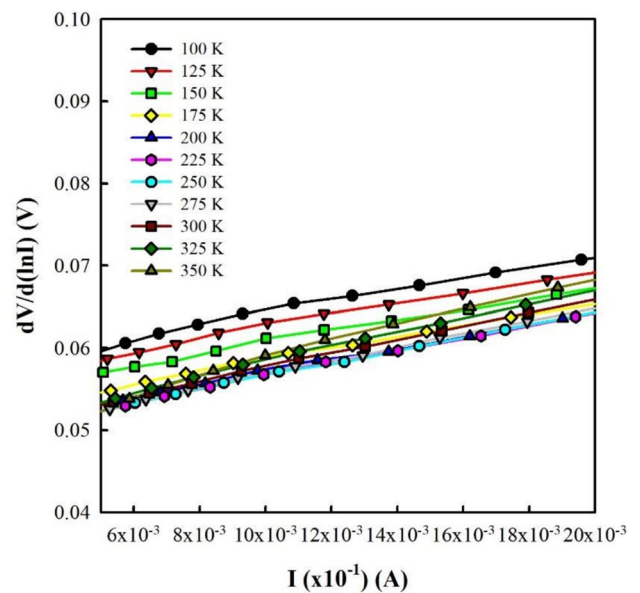


Fig. 5 $dV/d \ln(I)$ vs. I characteristics obtained from forward bias current–voltage measurements taken over a wide temperature range for the Au/PDI/n-Ge Schottky diode

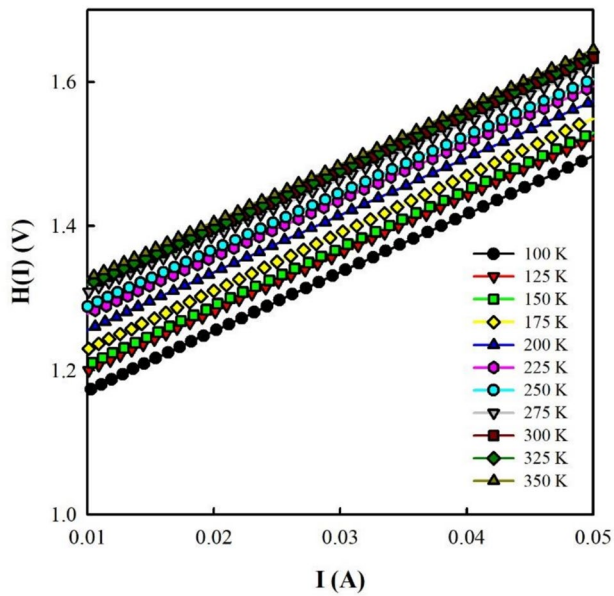


Fig. 6 $H(I)$ vs. I characteristics measured over a wide temperature range for the Au/PDI/n-Ge Schottky diode

found from the current–voltage characteristics. These values found using different methods are in good agreement with each other. The R_s values calculated from the Cheung plots decrease as the temperature increases. This decrease observed in R_s value is attributed to the decrease in n with temperature rise and besides to the rising of free carrier concentration observed at high temperatures.

The increase in series resistance observed at lower temperatures can primarily be attributed to the reduction of thermally activated carriers in the n-Ge substrate. As the temperature decreases, partial carrier freeze-out reduces the free carrier concentration and consequently increases the bulk resistance of the semiconductor. In addition, the PDI interlayer may introduce an additional temperature-dependent transport limitation through thermally activated hopping between localized states. Therefore, while both mechanisms may contribute to the observed behavior, the dominant contribution is considered to arise from carrier freeze-out in the n-Ge region, whereas hopping-limited transport within the PDI layer acts as a secondary effect.

3.3 Barrier inhomogeneity

It should be emphasized that the barrier heights obtained directly from conventional thermionic emission analysis represent apparent barrier heights rather than the intrinsic physical barrier of the interface. The relatively large ideality factors observed in the present study indicate significant deviations from ideal thermionic emission behavior, suggesting the presence of interface-related non-idealities and barrier inhomogeneity. Therefore, the apparent barrier

parameters extracted from the forward I–V characteristics were further analyzed using Gaussian distribution and modified Richardson approaches in order to obtain more representative intrinsic barrier parameters of the Au/PDI/n-Ge interface. The observed temperature dependence of the barrier height and ideality factor suggests that the barrier height is not uniform across the interface [55, 61, 63]. To explain this behavior, a Gaussian distribution of the barrier heights is assumed due to the barrier inhomogeneity at the metal–semiconductor interface [65]. This inhomogeneity can be attributed to presence of interfacial layers.

To examine this barrier inhomogeneity in more detail, the barrier height was plotted against the ideality factor. As demonstrated in Fig. 7, the dual linear correlation observed between the ideality factor and barrier height validates the existence of inhomogeneities at the interface. For the 100–150 K and 175–350 K temperature intervals, the mean barrier height values (ϕ_b) were determined from the slopes as 0.496 eV and 0.896 eV, respectively, for the Au/PDI/n-Ge interface. Furthermore, the existence of two separate linear trends supports the double Gaussian distribution model [63], suggesting that the barrier height distribution is not uniform across the entire Au/PDI/n-Ge junction.

The following equation is employed to produce the traditional and modified Richardson plots, providing an alternative method to analyze the temperature-dependent characteristics of ϕ_b and A^* [64].

$$\ln \left(\frac{I_0}{T^2} \right) = \ln (AA^*) - \frac{q\phi_b}{kT} \quad (9)$$

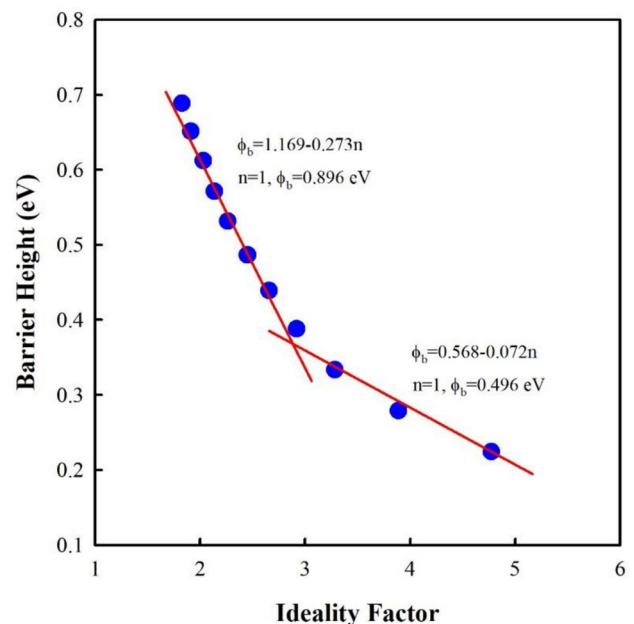


Fig. 7 The relationship between the n and ϕ_b for Au/PDI/n-Ge diode

Based on this equation, the conventional Richardson plot of $\ln(I_0/T^2)$ versus $1000/kT$ and the modified Richardson plot of $\ln(I_0/T^2)$ versus $1000/nkT$ are presented in Fig. 8. While the traditional plot exhibits a non-linear behavior due to the barrier inhomogeneity, the modified plot yields a high degree of linearity. From the slope and intercept of the linear fit in Fig. 8, barrier height (ϕ_b) and the Richardson constant (A^*) were determined. For the temperature interval of 225–350 K, the extracted parameters from the Richardson plots are as follows: $A^* = 1.5 \times 10^{-6} \text{ A cm}^{-2} \text{ K}^{-2}$ and $\phi_b = 0.136 \text{ eV}$ for fit 1, and $A^* = 1.86 \times 10^{-2} \text{ A cm}^{-2} \text{ K}^{-2}$ and $\phi_b = 0.768 \text{ eV}$ for the modified fit 2.

The observed temperature dependence of the ideality factor and barrier height is a characteristic feature of real Schottky junctions and is often associated with contact inhomogeneity. Many studies in the literature have reported such temperature sensitive variations in diodes due to barrier inhomogeneity on the sample surface and the Gaussian distribution (GD) of barrier heights [12, 16, 19]. In this model, the barrier height is considered as a continuous spatial distribution formed by regional patches with different heights, rather than a single value along the interface. This statistical approach, defined through the parameters of mean barrier height ($\bar{\phi}_{b0}$) and standard deviation (σ_{s0}), explains the temperature-dependent variation of the apparent barrier height (ϕ_{ap}) with the following equations [12], [16], [19]:

$$\phi_{ap} = \bar{\phi}_{b0} (T = 0) - \frac{q\sigma_{s0}^2}{2kT} \quad (10)$$

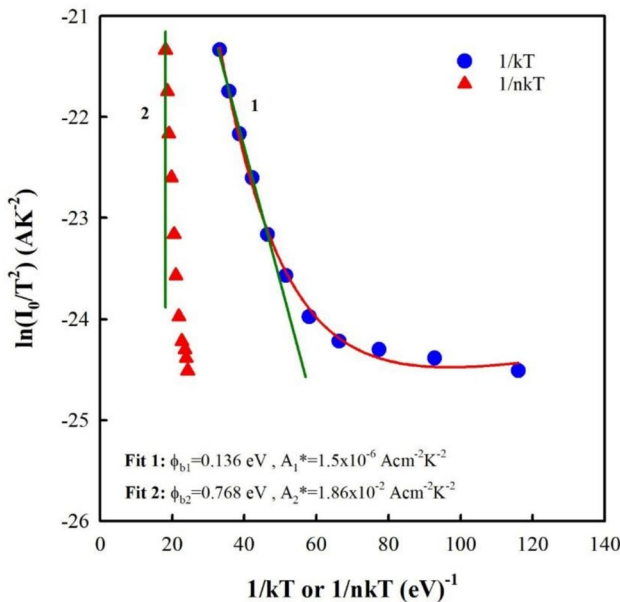


Fig. 8 Richardson plots of $\ln(I_0/T^2)$ vs $1/kT$ or $1/nkT$ for Au/PDI/n-Ge diode

When examining the temperature dependence of the apparent barrier height (ϕ_{ap}) presented in Fig. 9, the existence of two distinct linear regions becomes evident. This confirms the presence of a double-Gaussian distribution at the contact area. Based on the analyses, the average barrier height is calculated as 0.590 eV ($\sigma_{s01} = 0.080 \text{ V}$) in the 100–175 K range, and as 1.004 eV ($\sigma_{s02} = 0.141 \text{ V}$) in the 200–350 K range. It should be emphasized that the apparent barrier height obtained from conventional thermionic emission analysis at 350 K (approximately 0.69 eV) and the mean barrier heights extracted from the double-Gaussian distribution model represent different physical quantities. The value of 0.69 eV corresponds to the effective barrier height governing thermionic emission, whereas the higher values obtained from the Gaussian distribution analysis are statistical fitting parameters describing the spatial distribution of barrier heights across the interface rather than direct semiconductor band-edge energies. Therefore, these quantities should not be directly compared with the bandgap energy of Ge. Similarly to the barrier height analysis, two linear regions are observed here as well, indicating the presence of a double Gaussian distribution in the contact area.

Although the bandgap of Ge exhibits a temperature dependence, the change in bandgap over the investigated temperature range is relatively small compared with the observed variation in the apparent barrier height. Therefore, the temperature-dependent behavior of the extracted barrier parameters is primarily attributed to barrier inhomogeneity, interface states, and transport-related effects

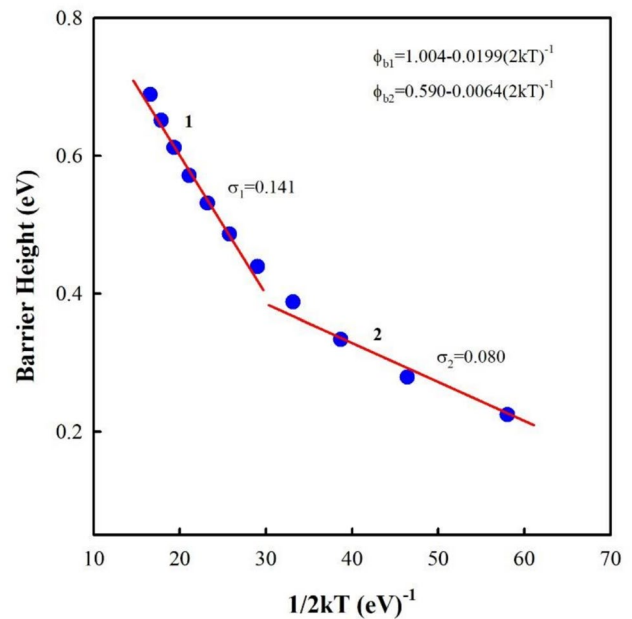


Fig. 9 Variation of the apparent barrier height as a function of $1/2kT$ for Au/PDI/n-Ge diode

rather than to the intrinsic temperature dependence of the Ge band structure.

By eliminating the effects of interface inhomogeneity, a more reliable value of the Richardson constant (A^*) can be obtained through the modified Richardson plot. To further confirm the barrier height value of the interfacial contact, the conventional Richardson plot expressed in Fig. 8 and by Eq. 9 can be transformed into the following equation by considering the Gaussian distribution of the barrier height [12], [16], [19]:

$$\ln \left(\frac{I_0}{T^2} \right) - \left(\frac{q^2 \sigma_{s0}^2}{2k^2 T^2} \right) = \ln(AA^*) - \frac{q\bar{\phi}_{b0}}{kT} \quad (11)$$

Figure 10 shows the modified plots of $\ln \left(\frac{I_0}{T^2} \right) - \left(\frac{q^2 \sigma_{s0}^2}{2k^2 T^2} \right)$ versus $1/kT$ for the Au/PDI/n-Ge Schottky diode. As observed from Fig. 10, the diode exhibits good linearity in the temperature ranges of 100–175 K and 200–350 K. From the intercepts and slopes of the two straight line regions, two sets of values for the Richardson constant A^* and the average barrier height $\bar{\phi}_{b0}$ are obtained: $A^* = 225.86 \text{ A cm}^{-2} \text{ K}^{-2}$ and $\bar{\phi}_{b0} = 0.593 \text{ eV}$ and in the 100–175 K range, and $A^* = 200.32 \text{ A cm}^{-2} \text{ K}^{-2}$ and $\bar{\phi}_{b0} = 1.010 \text{ eV}$ in the 200–350 K range, respectively. Moreover, the value of $A^* = 200.32 \text{ A cm}^{-2} \text{ K}^{-2}$ obtained for the 200–350 K range is close to the well-known value of $140 \text{ A cm}^{-2} \text{ K}^{-2}$ for electrons in n-Ge.

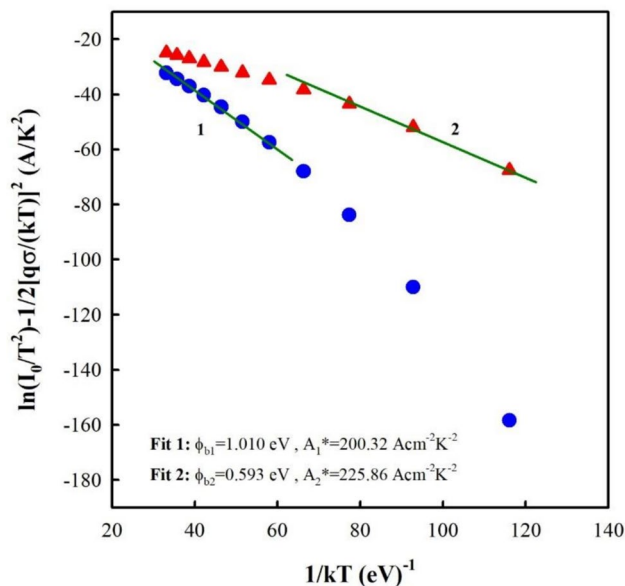


Fig. 10 Modified Richardson $\ln(I_0/T^2) - (q^2 \sigma_{s0}^2 / 2k^2 T^2)$ vs $1/kT$ curve for the Au/PDI/n-Ge structure based on Gaussian distribution of the barrier height

3.4 Series and shunt resistance analysis from R_i -V characteristics

The series resistance (R_s) and shunt resistance (R_{sh}) are among the most important parameters determining the performance of Schottky barrier diodes, as they directly influence the current transport mechanisms, leakage behavior, and rectification characteristics. These parameters were extracted from the differential resistance–voltage ($R_i - V$) characteristics obtained from the I–V measurements at different temperatures.

The differential resistance is defined as [52]:

$$R_i = \frac{dV}{dI} \quad (12)$$

and provides detailed insight into the voltage-dependent resistive behavior of the device.

The voltage-dependent differential resistance ($R_i - V$) characteristics of the fabricated diode were systematically analyzed over a wide temperature range (100–350 K) in order to gain deeper insight into the resistive behavior and underlying current transport mechanisms and are presented in Fig. 11. The differential resistance, defined as $R_i = dV/dI$, provides a sensitive probe for distinguishing between different conduction regimes and evaluating the influence of temperature on charge transport.

As clearly observed from the $R_i - V$ curves, the device exhibits a strong voltage- and temperature-dependent resistive behavior. At low bias voltages, the differential resistance values are remarkably high for all temperatures, particularly at lower temperatures (50–150 K). This behavior can be attributed to the limited carrier injection and reduced carrier mobility caused by insufficient thermal energy. In this region, the conduction is strongly suppressed, and the current is dominated by leakage pathways associated with high shunt resistance (R_{sh}). The pronounced increase in R_i at low temperatures further supports the presence of carrier freeze-out effects and reduced participation of thermally activated carriers.

With increasing applied voltage, a rapid decrease in R_i is observed for all temperatures. This transition region corresponds to the onset of significant carrier injection into the semiconductor, where the device begins to deviate from purely resistive behavior. The sharp decline in R_i indicates that the transport mechanism evolves from an ohmic conduction regime toward a space-charge-limited current (SCLC) regime. This behavior is consistent with the progressive filling of trap states and the increasing dominance of injected carriers over thermally generated carriers.

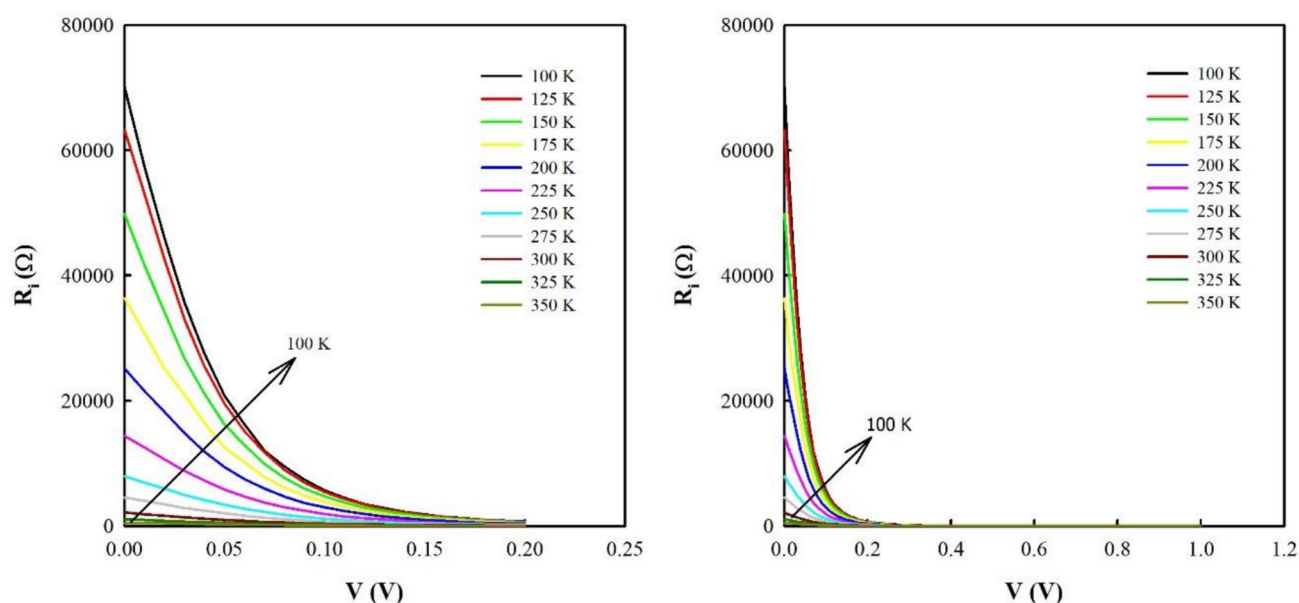


Fig. 11 Variation of R_i plotted against V for the Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

At higher bias voltages, the R_i values approach a minimum and become relatively less sensitive to voltage variations. In this region, the conduction is mainly governed by the series resistance (R_s) of the device. The nearly constant R_i behavior at high voltages indicates that the bulk resistance of the semiconductor and the contact resistances dominate the overall device response. This observation is a clear signature of the transition to a series resistance-limited conduction regime.

The temperature dependence of the $R_i - V$ characteristics reveals additional important insights. As the temperature increases from 100 to 350 K, the overall R_i values significantly decrease across the entire voltage range. This trend reflects the enhanced carrier concentration and mobility due to thermal activation. The increased thermal energy facilitates carrier excitation from trap states and reduces the potential barriers at the metal/semiconductor interface, leading to improved conductivity.

The extracted series resistance (R_s) values remain nearly constant (~ 8.6 – 8.8Ω) over the investigated temperature range, indicating that the series resistance is primarily governed by temperature-independent factors such as the bulk semiconductor resistance and the quality of ohmic contacts. This weak temperature dependence suggests that the contact interfaces are well-formed and stable, with minimal influence from thermally activated processes.

In contrast, the shunt resistance (R_{sh}) exhibits a strong temperature dependence, decreasing drastically with increasing temperature. Specifically, R_{sh} decreases from the order of $\sim 10^4 \Omega$ at low temperatures to $\sim 10^2 \Omega$ at higher temperatures. This pronounced reduction in R_{sh}

is indicative of increased leakage currents through defect states, interface traps, and possible localized conduction paths. The activation of these leakage channels at elevated temperatures significantly contributes to the deviation from ideal diode behavior.

The combined analysis of $R_i - V$ characteristics and extracted resistance parameters strongly supports the presence of multiple current transport mechanisms within the device. At low voltages and low temperatures, the conduction is dominated by high-resistance leakage pathways. As the voltage increases, the device transitions into a regime governed by injected carriers, corresponding to SCLC behavior. At higher voltages and temperatures, trap-controlled conduction (TCLC) and series resistance effects become dominant. Therefore, the $R_i - V$ analysis provides crucial experimental evidence for the voltage- and temperature-driven evolution of conduction mechanisms in the studied Schottky structure.

3.5 Determination of barrier height and series resistance using Norde method

In order to obtain reliable values of the barrier height and series resistance, the forward bias I-V characteristics were further analyzed using the Norde method. This method is particularly advantageous for Schottky-type structures exhibiting non-ideal behavior, as it minimizes the influence of series resistance and provides more accurate parameter extraction compared to conventional thermionic emission analysis. According to Norde's approach, the auxiliary function $F(V)$ is defined as [1], [42]:

$$F(V) = \frac{V}{\gamma} - \frac{kT}{q} \ln \left(\frac{I(V)}{AA^*T^2} \right) \quad (13)$$

where γ is an integer greater than the ideality factor ($\gamma > n$), A is the diode area, and A^* is the Richardson constant. The barrier height ϕ_b and series resistance R_s can then be determined from the minimum point of the $F(V) - V$ curve using the following relations:

$$\phi_b = F(V_{min}) + \frac{V_{min}}{\gamma} - \frac{kT}{q}; R_s = \frac{kT}{qI_{min}} (\gamma - n) \quad (14)$$

where V_{min} and I_{min} correspond to the voltage and current at the minimum point of the $F(V)$ curve.

The temperature-dependent $F(V) - V$ plots are presented in Fig. 12. As clearly observed, all curves exhibit a well-defined minimum point, which shifts systematically with temperature. With increasing temperature, the minimum of the $F(V)$ function moves toward higher voltage values and higher function values. This behavior reflects the strong influence of temperature on the barrier properties and carrier transport processes. The extracted barrier height and series resistance values as a function of temperature are summarized in Table 1. The barrier height (ϕ_b) increases from approximately 0.216 eV at 100 K to 0.687 eV at 350 K, while the series resistance R_s decreases significantly from about 88 Ω to 6.5 Ω over the same temperature range.

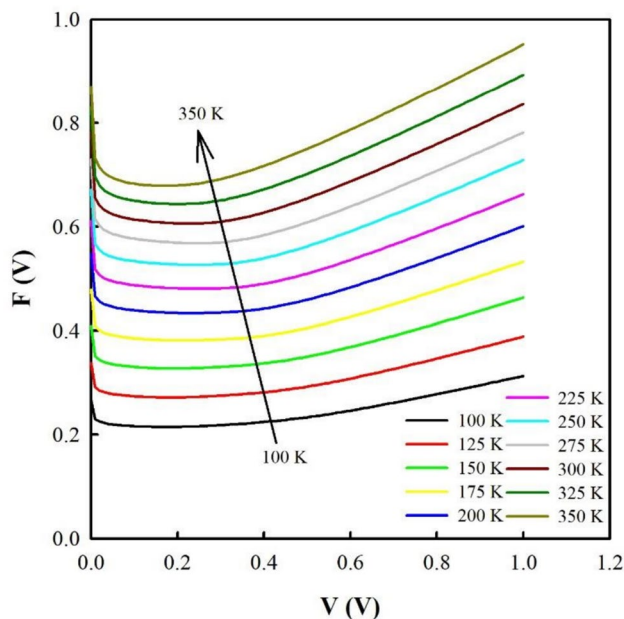


Fig. 12 Norde plot of Au/PDI/n-Ge Schottky barrier diodes at the measured temperatures

3.6 Analysis of current transport mechanisms using $\ln(I) - \ln(V)$ characteristics

To further elucidate the dominant current transport mechanisms in the fabricated device, the forward-bias current–voltage characteristics were analyzed on a double logarithmic scale ($\ln I - \ln V$) over the entire temperature range (100–350 K). This method is widely recognized as a powerful tool for distinguishing between different conduction regimes based on the slope of the $\ln(I)$ versus $\ln(V)$ plots. According to the power-law relationship [60]:

$$I \propto V^m \quad (15)$$

where m represents the slope of the $\ln(I) - \ln(V)$ curve, different conduction mechanisms can be identified depending on the value of m . Figure 13 shows the $\ln(I) - \ln(V)$ graphs, revealing three different voltage regions, each corresponding to a different dominant conduction mechanism.

In the low-voltage region (0.1–0.11 V), the slope values are approximately equal to unity ($m \approx 1$), indicating that the current increases linearly with applied voltage. This behavior corresponds to ohmic conduction, where thermally generated free carriers dominate the transport process. In this regime, the current is governed by the intrinsic carrier concentration, low electric field, and negligible space charge effects. The presence of ohmic conduction at low voltages confirms that the device exhibits a well-defined initial conduction region with minimal influence from trap states or

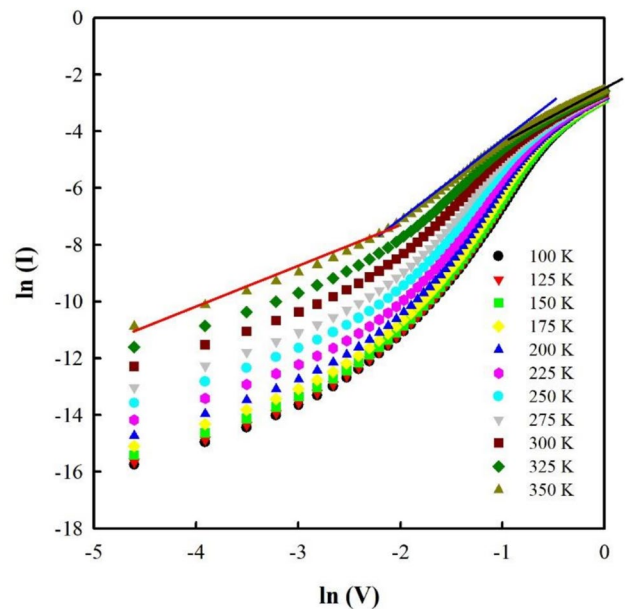


Fig. 13 Forward bias $\ln(I)$ vs. $\ln(V)$ plot of Au/PDI/n-Ge Schottky barrier diodes in the temperature range from 100 K–350 K

injected carriers. As the applied voltage increases (0.13–0.55 V), the slope values exceed unity ($m > 2$), indicating a transition to a trap-controlled conduction regime. This region corresponds to trap-controlled space-charge-limited current (T-SCLC), where the injected carriers begin to fill the localized trap states within the interfacial layer and semiconductor bulk. In this region, the traps significantly influence carrier transport, the injected carriers dominate over thermally generated carriers, and the current increases more rapidly with voltage. The existence of this region suggests a considerable density of trap states, which play a crucial role in shaping the electrical characteristics of the device. At higher voltages (0.56–1 V), the slope approaches a value close to 2 ($m \approx 2$), indicating that the conduction mechanism transitions to trap-free space-charge-limited current (SCLC). In this regime, the majority of trap states are filled, and the current is controlled by the space charge of injected carriers. This region is characterized by the strong carrier injection, dominance of space charge effects, and reduced influence of trap states. The observation of SCLC behavior confirms that the device reaches a high-injection regime where the current transport is governed by bulk properties rather than interface limitations.

Overall, the detailed $\ln(I)$ – $\ln(V)$ analysis demonstrates that the current transport in the device is governed by a multi-stage conduction process, evolving from ohmic conduction to trap-controlled SCLC and finally to trap-free SCLC with increasing voltage. This behavior highlights the critical role of trap states and their temperature-dependent activation in determining the electrical performance of the diode.

3.7 Reverse bias conduction mechanism analysis

In order to identify the dominant current transport mechanism under reverse bias conditions, the temperature-dependent current–voltage characteristics were analyzed using the $\ln(I_R)$ versus $V_R^{1/2}$ representation, as shown in Fig. 14. This method is widely used to distinguish between Schottky emission (SE) and Poole–Frenkel emission (PFE), both of which are electric field-assisted thermionic emission processes. Under reverse bias, the current can be expressed as [2], [46]:

$$I_R \propto \exp\left(\frac{\beta\sqrt{E}}{kT}\right) \quad (16)$$

For Schottky emission, which originates from barrier lowering at the metal–semiconductor interface, the coefficient is given by:

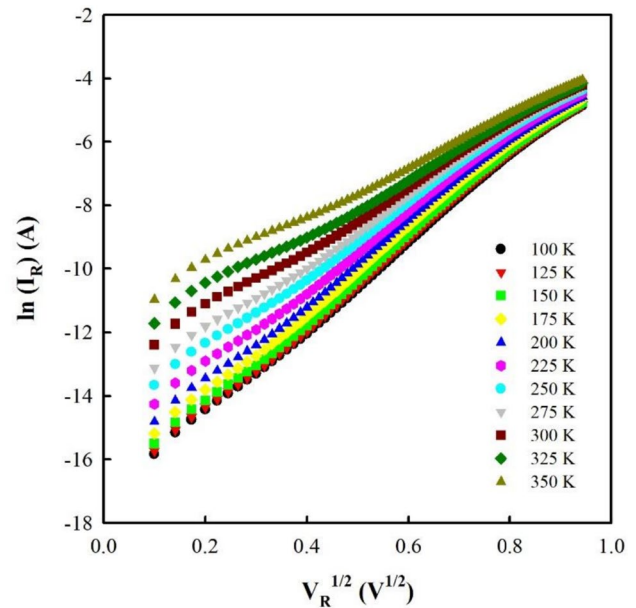


Fig. 14 Variation of $\ln I_R$ plotted against $V_R^{1/2}$ for the Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

$$\beta_{SE} = \sqrt{\frac{q^3}{4\pi\epsilon_0\epsilon_r}} \quad (17)$$

whereas for Poole–Frenkel emission, which involves field-assisted emission of trapped carriers from the bulk or interface states, the coefficient becomes:

$$\beta_{PFE} = 2\beta_{SE} \quad (18)$$

Using the dielectric constant of the PDI organic interlayer ($\epsilon_r = 3.4$), the theoretical values were calculated as $\beta_{SE} = 2.06 \times 10^{-5} \text{ eV m}^{1/2} \text{ V}^{-1/2}$ and $\beta_{PFE} = 4.11 \times 10^{-5} \text{ eV m}^{1/2} \text{ V}^{-1/2}$. As clearly seen in Fig. 14, the $\ln(I_R)$ versus $V_R^{1/2}$ plots exhibit a linear behavior over a wide voltage range for all temperatures between 100 and 350 K, indicating that the reverse current is governed by a field-assisted emission mechanism. The experimental β values extracted from the slopes of these plots are listed in Table 2 and vary between 1.19×10^{-5} and $2.61 \times 10^{-5} \text{ eV m}^{1/2} \text{ V}^{-1/2}$. It is observed that the β parameter increases with temperature up to approximately 300 K and then exhibits a slight decrease at higher temperatures.

A direct comparison between theoretical and experimental values reveals that the obtained β_{exp} values are in very close agreement with the theoretical Schottky coefficient, whereas they remain significantly lower than the Poole–Frenkel prediction. This result clearly indicates that the dominant reverse bias conduction mechanism is Schottky emission rather than Poole–Frenkel emission. The small deviations between experimental and theoretical Schottky

Table 2 Carrier transport parameters of Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

T (K)	$\mu_{\text{eff}} (\text{m}^2\text{V}^{-1}\text{s}^{-1}) (10^{-10})$	D (10^{-11})	τ (s) (10^{-8})	L_D (m) (10^{-9})	$\beta (\text{eV m}^{1/2}\text{V}^{-1/2}) (10^{-5})$
100	6.936	0.598	8.902	1.032	1.192
125	6.970	0.751	8.860	1.154	1.486
150	6.994	0.904	8.829	1.264	1.758
175	7.011	1.058	8.807	1.365	2.003
200	7.027	1.212	8.787	1.460	2.225
225	7.041	1.366	8.770	1.548	2.384
250	7.054	1.521	8.754	1.632	2.505
275	7.060	1.675	8.746	1.711	2.606
300	7.047	1.823	8.762	1.788	2.611
325	7.035	1.972	8.777	1.861	2.578
350	7.014	2.117	8.803	1.931	2.477

values, particularly at lower temperatures, can be attributed to several non-ideal effects such as barrier height inhomogeneity, interface state density, non-uniform electric field distribution, and image-force lowering variations. These factors lead to a slight dispersion in the effective barrier height and consequently influence the extracted β values.

Although the experimental β values are in closer agreement with the theoretical Schottky emission coefficient, the slight deviations observed especially at lower temperatures suggest that secondary conduction mechanisms may also contribute weakly to the reverse current. At low temperatures, reduced thermionic emission probability may enhance the relative contribution of localized trap-assisted transport pathways. Therefore, trap-assisted tunneling and/or weak Poole–Frenkel emission through interface or bulk trap states cannot be completely excluded in this region. Nevertheless, since the extracted β values remain much closer to the theoretical Schottky emission value than to the Poole–Frenkel value, Schottky emission is considered to be the dominant reverse-bias conduction mechanism in the Au/PDI/n-Ge diode.

The temperature dependence of the β parameter further supports this interpretation. The increase in β with temperature suggests that thermally activated carriers contribute more effectively to the emission process, reducing the influence of trap-assisted conduction. At higher temperatures, the slight deviation from the ideal trend may indicate the onset of additional leakage mechanisms or the partial contribution of trap-assisted tunneling processes. Furthermore, the reverse bias analysis is in strong agreement with the forward bias transport results discussed earlier. While the forward bias characteristics are dominated by ohmic conduction, trap-controlled space-charge-limited current (T-SCLC), and trap-free SCLC mechanisms, the reverse bias current is primarily controlled by Schottky emission. This asymmetry in conduction mechanisms highlights the dominant role of the

metal–semiconductor interface under reverse bias and confirms that bulk trap-controlled conduction plays a secondary role in this regime.

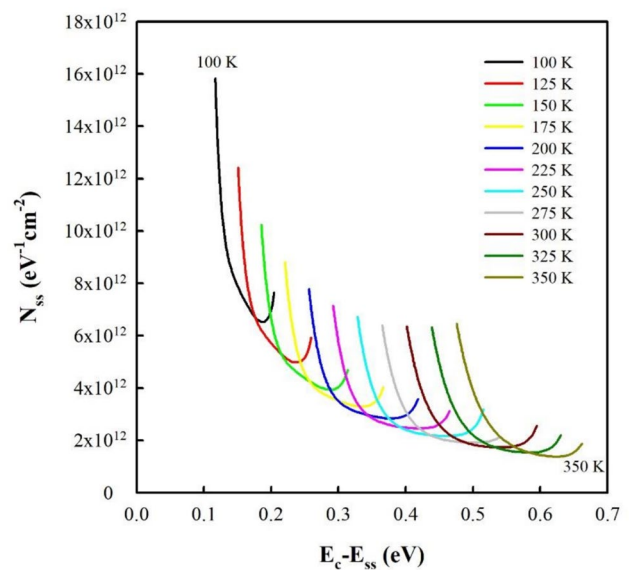
In conclusion, the close agreement between experimental and theoretical Schottky β values, together with the linear behavior of $\ln(I_R)$ versus $V_R^{1/2}$ plots shown in Fig. 14, provides strong evidence that the reverse current transport in the fabricated device is predominantly governed by Schottky emission. This finding demonstrates that the interface-controlled barrier lowering mechanism is the key factor determining the reverse bias characteristics of the diode.

3.8 Energy-dependent interface state density (N_{ss}) analysis

The energy distribution of interface state density (N_{ss}) as a function of $(E_c - E_{ss})$ was investigated using the forward bias I–V characteristics over the temperature range of 100–350 K, as shown in Fig. 15. The determination of N_{ss} is of critical importance for understanding the electronic quality of the metal/organic/semiconductor interface, as these states strongly influence carrier recombination, barrier height, and overall device performance.

The interface state density was calculated by considering the voltage-dependent ideality factor and barrier height using the following well-known relation [28], [62]:

$$N_{ss} = \frac{1}{qA} \left[\frac{\varepsilon_i}{\delta} (n - 1) - \frac{\varepsilon_s}{W_D} \right] \quad (19)$$

**Fig. 15** Variation of density of states (N_{ss}) plotted against $E_c - E_{ss}$ for the Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

where q is the electronic charge, A is the diode area, ε_i and ε_s are the permittivities of the interfacial layer and semiconductor, respectively, δ is the thickness of the interfacial layer, W_D is the depletion width, and n is the ideality factor. The energy level of the interface states relative to the conduction band edge was determined using [48]:

$$E_c - E_{ss} = q(\phi_b - V) \quad (20)$$

As clearly observed in Fig. 15, the interface state density exhibits a strong dependence on both energy and temperature. At low temperatures (100–175 K), the N_{ss} values are significantly higher, reaching the order of $10^{13} \text{ eV}^{-1} \text{ cm}^{-2}$, particularly near the conduction band edge (low $E_c - E_{ss}$ values). This behavior indicates a high density of localized trap states that are energetically close to the conduction band and actively participate in carrier trapping. As the temperature increases, a systematic decrease in is observed across the entire energy range. At higher temperatures (200–350 K), the interface state density reduces to the order of $10^{12} \text{ eV}^{-1} \text{ cm}^{-2}$, indicating a significant improvement in the effective interface quality.

Another important observation is that N_{ss} decreases exponentially with increasing $(E_c - E_{ss})$. This indicates that the interface states are densely distributed near the band edge and the deeper energy levels contain fewer active traps. This type of distribution is characteristic of exponentially distributed interface states, which is commonly observed in organic interlayer-based Schottky structures.

3.9 Carrier transport parameters derived from $J - V$ and $J - V^2$ characteristics

In order to gain deeper insight into the carrier transport dynamics of the fabricated device, the current density–voltage ($J - V$) and $J - V^2$ characteristics were analyzed over the temperature range of 100–350 K, as presented in Fig. 16 and 17. From these analyses, key transport parameters including the effective carrier mobility (μ_{eff}), diffusion coefficient (D), transit time (τ), and diffusion length (L_D) were extracted.

As observed in Fig. 16, the $J - V$ plots exhibit a systematic increase in current density with increasing temperature. This behavior indicates that carrier transport is strongly influenced by thermal activation, allowing charge carriers to overcome potential barriers and trap states more efficiently at higher temperatures. Under an applied electric field, the current density in the drift-dominated regime can be expressed as [4]:

$$J = \frac{9\varepsilon_r\varepsilon_0}{8\tau} \frac{V}{d} \quad (21)$$

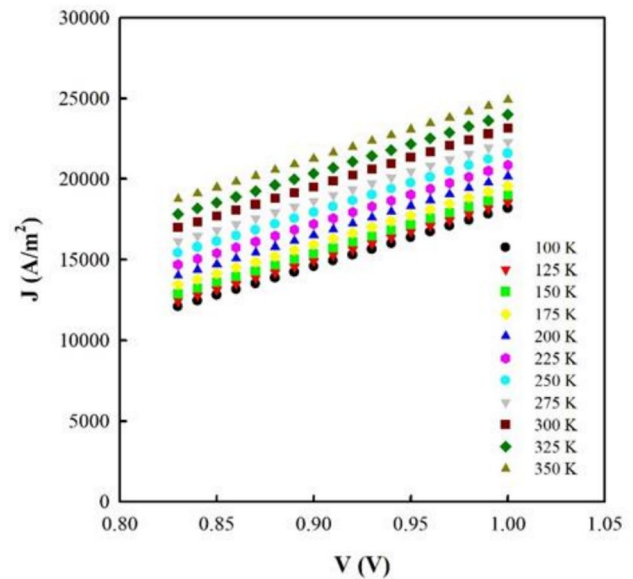


Fig. 16 Variation of $J - V$ plots for the Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

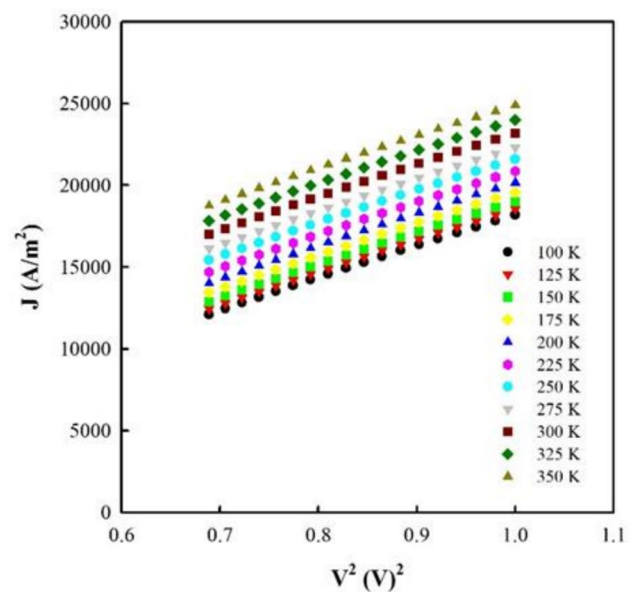


Fig. 17 Variation of $J - V^2$ plots for the Au/PDI/n-Ge diode in the temperature range from 100 to 350 K

Here, the transit time (τ) can be defined as the time required for carriers to cross the active layer under an applied electric field. The transit time (τ) values were calculated and found to be on the order of 10^{-8} s , decreasing slightly from $8.902 \times 10^{-8} \text{ s}$ at 100 K to $8.803 \times 10^{-8} \text{ s}$ at 350 K. This decrease indicates that carriers move faster through the device at higher temperatures, which is consistent with enhanced conduction observed in the $J - V$ plots.

The $J - V^2$ plots in Fig. 17 show a linear dependence at higher voltages for all temperatures, confirming that the

transport mechanism in this region is governed by space-charge-limited current (SCLC). According to the Mott–Gurney law [4], [45]:

$$J = \frac{9\mu_{eff}\varepsilon_r\varepsilon_0}{8} \frac{V^2}{d^3} \quad (22)$$

the effective carrier mobility (μ_{eff}) can be extracted from the slope of the linear region. The calculated μ_{eff} values vary between 6.936×10^{-10} and $7.060 \times 10^{-10} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$, showing only a weak dependence on temperature. This indicates that the transport process is not strongly limited by phonon scattering, but rather by structural disorder and localized states in the organic interlayer.

The diffusion coefficient (D) was calculated using the Einstein relation [4]:

$$D = \mu_{eff} \frac{kT}{q} \quad (23)$$

which shows a clear increase with temperature, ranging from 5.983×10^{-12} to $2.117 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$.

The diffusion length was then determined from [4]:

$$L_D = \sqrt{D\tau} \quad (24)$$

The calculated L_D values increase from 1.032×10^{-9} m to 1.931×10^{-9} m with increasing temperature, indicating improved carrier transport and reduced recombination at higher temperatures.

The combined analysis reveals that while the effective mobility remains nearly constant, the decrease in transit time and increase in diffusion length with temperature indicate enhanced carrier transport efficiency. This behavior can be attributed to the thermal activation of carriers, the reduced trap-assisted recombination decrease in interface state density (N_{ss}) improved conductivity (consistent with $R_i - V$ and Norde analyses). Thus, the transport mechanism evolves from trap-limited conduction at low temperatures toward more efficient SCLC-dominated transport at higher temperatures. As a result, the $J - V$ and $J - V^2$ analyses demonstrate that the device exhibits thermally assisted carrier transport, characterized by stable mobility, reduced transit time, and enhanced diffusion properties at elevated temperatures. These results highlight the crucial role of the organic interlayer in regulating charge transport and improving device performance.

4 Conclusion

The electrical behavior of the Au/PDI/n-Ge Schottky diode was systematically investigated over the temperature range of 100–350 K, revealing strong deviations from ideal thermionic emission (TE) due to pronounced barrier inhomogeneity. The zero-bias barrier height increased from ~ 0.224 eV to ~ 0.688 eV, while the series resistance decreased from $\sim 88 \Omega$ to $\sim 6.5 \Omega$, indicating thermally activated carrier transport. To quantitatively describe this non-ideal behavior, a double Gaussian distribution (GD) model was applied, assuming spatially distributed barrier heights across the interface. The analysis clearly revealed two distinct barrier distributions corresponding to different temperature regimes. The extracted Gaussian parameters are $\bar{\phi}_{b0} = 0.590$ eV and $\sigma_{s01} = 0.08$ V in the low-temperature region (100–175 K) and $\bar{\phi}_{b0} = 1.004$ eV and $\sigma_{s02} = 0.141$ V in the high-temperature region (200–350 K).

Contrary to conventional expectations, the smaller standard deviation at low temperatures does not indicate a more homogeneous interface, but rather reflects selective carrier transport through localized low-barrier patches. At low temperatures, carriers lack sufficient thermal energy to overcome higher barriers and therefore preferentially flow through these limited regions, resulting in an apparently narrower barrier distribution. As the temperature increases, carriers gain sufficient energy to surmount a wider range of barrier heights, revealing the true spatial inhomogeneity of the interface, which manifests as a larger σ value. This behavior is consistent with the Gaussian model and confirms that the interface consists of a distribution of barrier patches rather than a uniform barrier.

The modified Richardson analysis yielded an effective Richardson constant close to the theoretical value for n-Ge ($\sim 140 \text{ A cm}^{-2} \text{ K}^{-2}$), validating the GD-assisted TE model. In addition, current transport evolves from ohmic conduction to trap-controlled and trap-free space-charge-limited current (SCLC) regimes, while reverse bias conduction is governed by barrier-controlled emission mechanisms. Furthermore, the interface state density (N_{ss}) decreases with increasing temperature, indicating thermally activated reduction of trap states and improved interface quality.

In conclusion, the results demonstrate that the electrical performance of the Au/PDI/n-Ge Schottky diode is governed by temperature-dependent Gaussian-distributed barrier inhomogeneity, where the apparent narrowing of the distribution at low temperatures is a transport-limited effect,

and the broader distribution at higher temperatures reflects the intrinsic interface disorder. The PDI interlayer plays a crucial role in modulating this distribution and controlling the charge transport across the junction.

The significance of this work lies not in the development of a new transport model, but in demonstrating that the Au/PDI/n-Ge interface exhibits a distinct double-Gaussian barrier distribution together with multiple temperature-dependent transport regimes. These findings provide new insights into how PDI-based organic interlayers modify interfacial energetics and carrier transport in Ge-based Schottky devices, offering a promising route for the design of future low-cost organic/inorganic electronic structures. Moreover, The significance of this work lies not in achieving ideal diode characteristics, but in revealing how the PDI interlayer modifies barrier formation, interface inhomogeneity, and charge transport mechanisms in a Ge-based Schottky structure. These findings contribute to a better understanding of non-ideal transport processes at organic/inorganic interfaces.

The significance of this work lies not in the development of a new transport model, but in demonstrating that the Au/PDI/n-Ge interface exhibits a distinct double-Gaussian barrier distribution together with multiple temperature-dependent transport regimes. These findings provide new insights into how PDI-based organic interlayers modify interfacial energetics and carrier transport in Ge-based Schottky devices, offering a promising route for the design of future low-cost organic/inorganic electronic structures.

Although the extracted ideality factors indicate non-ideal transport behavior, the PDI interlayer effectively modifies the Au/n-Ge interface by increasing the barrier height and controlling interface-related transport processes. These characteristics make the Au/PDI/n-Ge structure a useful platform for studying interface engineering in organic/inorganic hybrid devices and may be beneficial for future sensor and photodetector applications where tunable interfacial properties are desired. Although the present study focuses on the temperature-dependent electrical behavior of the Au/PDI/n-Ge structure, future investigations involving long-term storage and environmental stability tests will be valuable for assessing the practical applicability and operational durability of PDI-based Schottky devices.

Acknowledgements The authors thank Dr. Mahmut Kuş from Konya Technical University for synthesized the PDI organic compound.

Author contributions Z.K.: Investigation, Writing – review & editing; Ü.A.: Writing – original draft, Formal Analysis; N.T.: Data curation, Writing – review & editing; Ö.F.Y.: Data collection, Methodology. All authors read and approved the final manuscript.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). There is no funding for this study.

Data availability All data generated and analyzed during this study are included in this publication.

Declarations

Competing interests The authors declare no competing interests.

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